

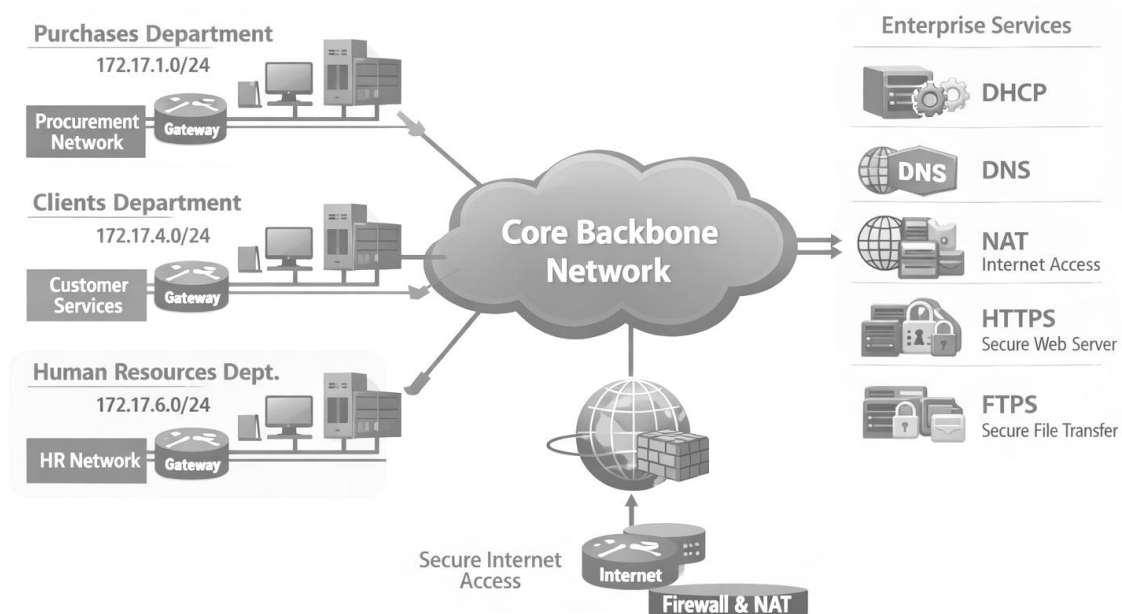
	Network administration	2025/2026
	Project N°4	

Objective:
Designing and administering a secure internal network with dedicated services (e.g., DHCP, DNS, NAT, HTTPS, FTPS)

ALMASSAR Company plans to design and deploy a secure, structured internal network to support its organizational and operational activities. The company is organized into three departments, each with distinct roles and communication requirements. The network must ensure controlled access to both internal and external resources:

- Purchases department: responsible for procurement/supplier interactions.
- Clients department: responsible for customer services and client-related operations.
- Human resources department: responsible for employee data and internal administration.

Each department must operate on its own dedicated network segment in order to isolate traffic, reduce congestion, and improve overall network performance. Communication within a department must remain local, while communication between departments must be controlled and routed through the core infrastructure. To interconnect these departmental networks, the enterprise plans to deploy a central backbone network that acts as the core of the infrastructure. This backbone will allow inter-department communication while preserving logical separation between departments. Each department will be connected to the backbone through a dedicated gateway, ensuring that all inter-department traffic is properly managed.



In addition to basic connectivity, the enterprise requires several essential network services to support daily operations and external communication:

- **DHCP:** the enterprise requires automatic IP address assignment for end-user devices.
- **DNS:** the network must include a name resolution service to allow users and systems to access internal and external resources using human-readable names rather than IP addresses.
- **NAT:** the enterprise requires controlled access to the Internet for internal users. So, a mechanism to translate internal private addresses into publicly routable addresses is required.
- **HTTPS:** the enterprise plans to host internal and/or external web applications. These services must be accessible through a secure web server using HTTPS.
- **FTPS:** to support secure exchange of documents with partners and internal users, the enterprise requires a secure file transfer service based on encrypted protocols.

The project can be implemented using any of the three platforms, **Linux network namespaces**, **GNS3**, or **EVE-NG**.

Submission Dates (Platform)

- Final document (maximum 10 pages)
- The report, source code, and test cases/scripts must be submitted in a single archive file (lastname-firstname-Project4.zip/gz).